

CS536 Science of Programming
Fall 2025
Assignment 5 Sample Solution Sketches

1. We can give the following formal proof.

1 $\{p \wedge B\} S_1 \{q_1\}$	premise
2 $\{p \wedge \neg B\} S_2 \{q_2\}$	premise
3 $\{(B \rightarrow p \wedge B) \wedge (\neg B \rightarrow p \wedge \neg B)\} \text{ if } B \text{ then } S_1 \text{ else } S_2 \text{ fi } \{q_1 \vee q_2\}$	if-else 1,2
4 $(B \rightarrow p \wedge B) \wedge (\neg B \rightarrow p \wedge \neg B) \Leftrightarrow (B \wedge (p \wedge B)) \vee (\neg B \wedge (p \wedge \neg B))$	predicate logic
5 $(B \wedge (p \wedge B)) \vee (\neg B \wedge (p \wedge \neg B)) \Leftrightarrow p$	predicate logic
$\# LHS \Leftrightarrow p \wedge B \vee p \wedge \neg B \Leftrightarrow p \wedge (B \vee \neg B) \Leftrightarrow p$	
6 $\{p\} \text{ if } B \text{ then } S_1 \text{ else } S_2 \text{ fi } \{q_1 \vee q_2\}$	strengthen precondition 5,3

2. $p_1 : x = 2^k \wedge k \leq n \wedge k < n$ which is logically equivalent to $x = 2^k \wedge k < n$

$p_2 : x_0 = 2^k \wedge k < n \wedge x = x_0 * 2$

R_1 : forward assignment

$p_3 : x_0 = 2^{k_0} \wedge k_0 < n \wedge x = x_0 * 2 \wedge k = k_0 + 1$

R_2 : forward assignment

R_3 : sequence 1,2

R_4 : weaken postcondition 3,4

$p_4 : p \wedge k \geq n$

R_5 : loop 5

3. R_1 : forward assignment

R_2 : forward assignment

R_3 : weaken postcondition 2,3

R_4 : sequence 1,4

R_5 : backward assignment

R_6 : backward assignment

R_7 : strengthen precondition 8,7

R_8 : sequence 9,6

R_9 : loop 10

R_{10} : sequence 5,11

R_{11} : weaken postcondition 12,13

4. (a) Here is the full proof outline.

```

{ $n > 0$ }  $k := n - 1$ ; { $n > 0 \wedge k = n - 1$ }  $x := n$ ; { $n > 0 \wedge k = n - 1 \wedge x = n$ }
{inv  $p \equiv 1 \leq k \leq n \wedge x = fac(x)/fac(k)$ }
while  $k > 1$  do
  { $p \wedge k > 1$ }
    { $p [x * k/x][k - 1/k]$ }  $k := k - 1$ ; { $p [x * k/x]$ }  $x := x * k$  { $p$ }
od
{ $p \wedge k \leq 1$ } { $x = fac(n)$ }

```

- (b) Here is the minimal proof outline.

```

{ $n > 0$ }  $k := n - 1$ ;  $x := n$ ;
{inv  $p \equiv 1 \leq k \leq n \wedge x = fac(n)/fac(k)$ }
while  $k > 1$  do
   $k := k - 1$ ;  $x := x * k$ 
od
{ $x = fac(n)$ }

```

5. Here is the full proof outline under partial correctness with the Forward Assignment Axiom used for all assignment statements.

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{ $y \geq 1$ }  $x := 0$ ; { $y \geq 1 \wedge x = 0$ }  $r := 1$ ; { $y \geq 1 \wedge x = 0 \wedge r = 1$ }
{inv  $p \equiv 1 \leq r = 2^x \leq y$ }
while  $2 * r \leq y$  do
  { $p \wedge 2 * r \leq y$ }
     $r := 2 * r$ ; { $1 \leq r_0 = 2^x \leq y \wedge 2 * r_0 \leq y \wedge r = 2 * r_0$ }  $x := x + 1$ 
    { $p_1 \equiv 1 \leq r_0 = 2^x \leq y \wedge 2 * r_0 \leq y \wedge r = 2 * r_0 \wedge x = x_0 + 1$ }
  { $p$ }
od
{ $p \wedge 2 * r > y$ } { $r = 2^x \leq y < 2^{x+1}$ }

```

We need to prove the following logic implication to finish the proof:

- With $y \geq 1 \wedge x = 0 \wedge r = 1$, we have the loop invariant p satisfied and we can start the loop. As an aside, $1 \leq r = 2^x \leq y \Leftrightarrow 1 \leq r \wedge r = 2^x \wedge 2^x \leq y$.
- $p_1 \Rightarrow p$. Since $1 \leq r_0 = 2^{x_0}$ and $r = 2 * r_0 \wedge x = x_0 + 1$, so $1 \leq r = 2^x$. Since $2 * r_0 \leq y$ and $r = 2 * r_0$, so $r \leq y$.
- $p \wedge 2 * r > y \Rightarrow r = 2^x \leq y < 2^{x+1}$. Since $p \equiv 1 \leq r = 2^x \leq y$, so $r = 2^x \leq y$. Since $2 * r > y$, so $y < 2^{x+1}$.

6. Here is one of the possible formal proofs for the full proof outline created in question 5.

1 $\{y \geq 1\} \ x := 0 \ \{y \geq 1 \wedge x = 0\}$	forward assignment
2 $\{y \geq 1 \wedge x = 0\} \ r := 1 \ \{y \geq 1 \wedge x = 0 \wedge r = 1\}$	forward assignment
3 $\{y \geq 1\} \ x := 0; \ r := 1 \ \{y \geq 1 \wedge x = 0 \wedge r = 1\}$	sequence 1,2
4 $y \geq 1 \wedge x = 0 \wedge r = 1 \Rightarrow p$	predicate logic
$\# \ p \equiv 1 \leq r = 2^\wedge x \leq y$	
5 $\{y \geq 1\} \ x := 0; \ r := 1 \ \{p\}$	weaken postcondition 3,4
6 $\{p \wedge 2 * r \leq y\} \ r := 2 * r \ \{1 \leq r_0 = 2^\wedge x \leq y \wedge 2 * r_0 \leq y \wedge r = 2 * r_0\}$	forward assignment
7 $\{1 \leq r_0 = 2^\wedge x \leq y \wedge 2 * r_0 \leq y \wedge r = 2 * r_0\} \ x := x + 1 \ \{p_1\}$	forward assignment
$\# \ p_1 \equiv 1 \leq r_0 = 2^\wedge x_0 \leq y \wedge 2 * r_0 \leq y \wedge r = 2 * r_0 \wedge x = x_0 + 1$	
8 $\{p \wedge 2 * r \leq y\} \ r := 2 * r; \ x := x + 1 \ \{p_1\}$	sequence 6,7
9 $p_1 \Rightarrow p$	predicate logic
10 $\{p \wedge 2 * r \leq y\} \ r := 2 * r; \ x := x + 1 \ \{p\}$	weaken postcondition 8,9
11 inv $p\} \ W \ \{p \wedge 2 * r > y\}$	loop 10
$\# \ W \equiv \textbf{while } 2 * r \leq y \textbf{ do } r := 2 * r; \ x := x + 1 \textbf{ od}$	
12 $\{y \geq 1\} \ x := 0; \ r := 1; W \ \{p \wedge 2 * r > y\}$	sequence 5,11
13 $p \wedge 2 * r > y \Rightarrow r = 2^\wedge x \leq y < 2^\wedge(x + 1)$	predicate logic
14 $\{y \geq 1\} \ x := 0; \ r := 1; W \ \{r = 2^\wedge x \leq y < 2^\wedge(x + 1)\}$	weaken postcondition 13,14

7. Here is one of the possible full proof outline for this triple under total correctness.

```

{p}
if  $\text{sqrt}(x) > y$  then
     $\{b[x - y] = y \wedge 0 \leq (x - y) < \text{size}(b)\} \ x := b[x - y] \ \{x = y\}$ 
else
     $\{x = b[y - x] \wedge 0 \leq (y - x) < \text{size}(b)\} \ y := b[y - x] \ \{x = y\}$ 
fi
{x = y}

```

Here, precondition $p \equiv (x \geq 0) \wedge (\text{sqrt}(x) > y \rightarrow b[x - y] = y \wedge 0 \leq (x - y) < \text{size}(b)) \wedge (\text{sqrt}(x) \leq y \rightarrow x = b[y - x] \wedge 0 \leq (y - x) < \text{size}(b))$. It is calculated using Conditional Rule 2 (total correctness version).

8. To obtain a safe precondition, let us start by calculating the domain of the statement.

$$\begin{aligned}
D(x := x * y; x := 1/x) &\equiv D(x := x * y) \wedge wp(x := x * y, D(x := 1/x)) \\
&\equiv T \wedge wp(x := x * y, x \neq 0) \\
&\Leftrightarrow x * y \neq 0
\end{aligned}$$

Here is a possible full proof outline for this triple under total correctness. We used

the Forward Assignment Axiom to prove both assignment statements in the program.

$$\{sqrt(x) \leq y \wedge x \geq 0 \wedge x * y \neq 0\}$$

$$x := x * y; \{sqrt(x_0) \leq y \wedge x_0 \geq 0 \wedge x_0 * y \neq 0 \wedge x = x_0 * y\}$$

$$x := 1/x \{sqrt(x_0) \leq y \wedge x_0 \geq 0 \wedge x_0 * y \neq 0 \wedge x_1 = x_0 * y \wedge x = 1/x_1\}$$